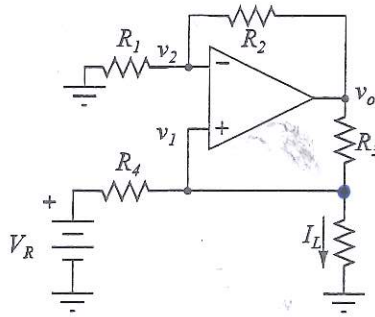


2014-2015 Fall Semester
EEE-211 Final Exam
Solutions

Question 1 [25 Points]:



For the circuit above, the OPAMP is not saturated.

- (a) [15 points] Find the current I_L in terms of the given parameters (V_R , v_o , R_1 , R_2 , R_3 , R_4). Simplify the expression as much as you can.

$$v_2 = v_o \cdot \frac{R_1}{R_1 + R_2} \quad (\text{voltage divider})$$

$$I_L + \frac{v_1 - V_R}{R_4} + \frac{v_1 - v_o}{R_3} = 0 \quad (\text{KCL at node } v_1)$$

$$\begin{aligned} I_L &= \frac{V_R}{R_4} - \frac{(R_3 + R_4)}{R_3 R_4} \cdot v_1 + \frac{v_o}{R_3} \quad \left. \begin{array}{l} \text{using } v_1 = v_2 = v_o \frac{R_1}{R_1 + R_2} \end{array} \right\} \\ &= \frac{V_R}{R_4} - \frac{(R_3 + R_4)}{R_3 \cdot R_4} \cdot \frac{R_1}{R_1 + R_2} v_o + \frac{v_o}{R_3} \\ &= \frac{V_R}{R_4} + \frac{v_o}{R_3 \cdot R_4 \cdot (R_1 + R_2)} \cdot (R_1 R_4 + R_2 R_4 - R_1 R_3 - R_1 R_4) \\ &= \frac{V_R}{R_4} + \frac{R_2 R_4 - R_1 R_3}{R_3 \cdot R_4 \cdot (R_1 + R_2)} \cdot v_o = \boxed{\frac{V_R}{R_4} + \frac{v_o}{R_1 + R_2} \left[\frac{R_2}{R_3} - \frac{R_1}{R_4} \right]} \end{aligned}$$

- (b) [10 points] We have $R_4 = R_1 R_3 / R_2$. Simplify the expression for I_L as much as you can.

$$I_L = \frac{V_R}{R_4} + \frac{v_o}{R_1 + R_2} \underbrace{\left[\frac{R_2}{R_3} - \frac{R_1}{R_1 \cdot R_3 / R_2} \right]}_{\text{zero!}}$$

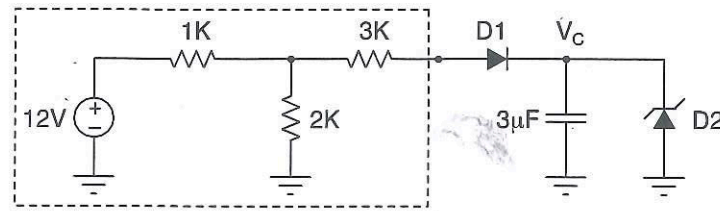
$$\boxed{I_L = \frac{V_R}{R_4}}$$

$$\text{OR } I_L = \frac{R_2}{R_1 R_3} V_R$$

↓
(comment: if our load is the resistor with I_L passing through it, this circuit is like a current source. The current amplitude does not depend on the load.)

(a) $I_L = \frac{V_R}{R_4} + \frac{v_o}{R_1 + R_2} \left[\frac{R_2}{R_3} - \frac{R_1}{R_4} \right]$	(b) $I_L = \frac{V_R}{R_4}$
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Question 2 [25 Points]:



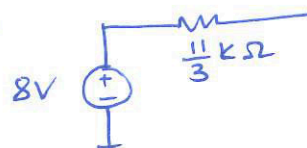
In the circuit above, the threshold voltage for diode D1 is $V_{ON} = 0.6V$, the breakdown voltage for Zener diode D2 is $V_Z = 5V$. For the capacitor, $V_C(0) = 0V$.

(a) [7 points] Find the Thevenin equivalent of the circuit inside the dashed box.

V_{TH} : $V_{TH} = 12 \cdot \frac{2k}{1k+2k} = \boxed{8V}$

R_{eq} : Kill all sources $R_{eq} = (1k // 2k) + 3k = \boxed{\frac{11}{3} k\Omega}$

So, Thevenin Equivalent is:



(b) [18 points] Find and plot $V_C(t)$ for $t > 0$. Clearly specify the state of each diode for all $t > 0$. Plot your result below. Mark all important voltage levels and time points.

- * Positive voltage across D1, so D1 is ON for all $t > 0$.
- * D2 is initially OFF, because it has zero volts across it.

* initially: $V_{init} = 0V$

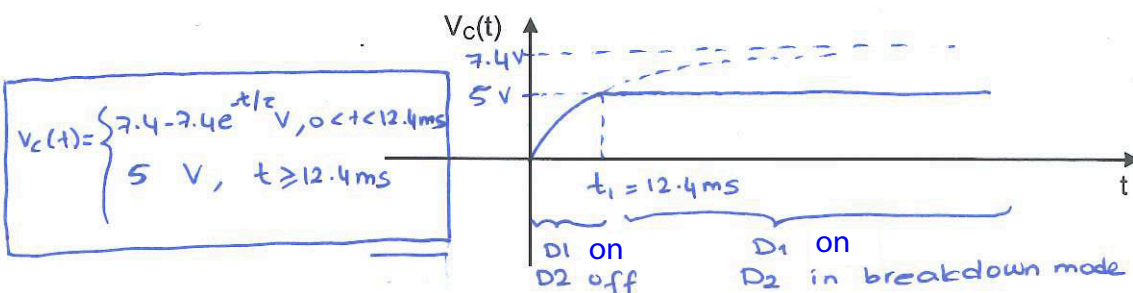
* $\tau = RC = \frac{11}{3} k \cdot 3\mu F = \boxed{11ms}$

* finally: Assume D2 remains OFF. Capacitor charges.

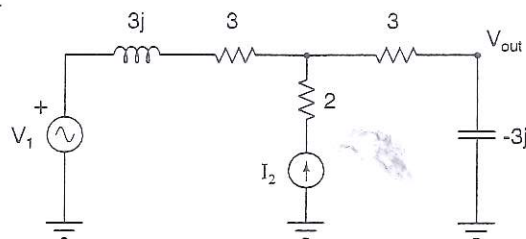
$V_{final} = 8 - 0.6 = 7.4V$

* So, $V_C(t) = V_{final} + (V_{init} - V_{final})e^{-t/\tau} = 7.4 - 7.4e^{-t/11ms} V$

* When V_C reaches 5V, Zener diodes goes into breakdown mode. $5 = 7.4 - 7.4e^{-t_1/11ms} \Rightarrow \boxed{t_1 = 12.4ms}$. After this, it sets the voltage to 5V, and capacitor cannot charge further.



Question 3 [25 points]:



In the circuit above, $v_1(t) = 6\sqrt{2} \cos(\omega t + \frac{\pi}{4})$ Volts, and $i_2(t) = 2 \sin(\omega t)$ Amps.

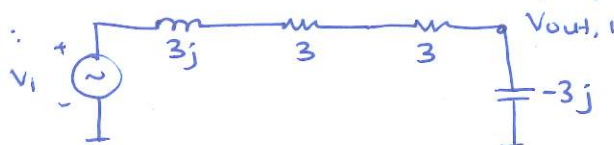
The impedances are given in units of Ohms at frequency ω .

$\rightarrow i_2(t) = 2 \cos(\omega t - \pi/2)$

(a) [20 points] Find V_{out} in phasor domain. Express it in polar form ($A \angle \theta$), with amplitude and phase both within two significant digits. Express the phase in radians.

* many ways to solve. I will use superposition.

* Due to V_1 :

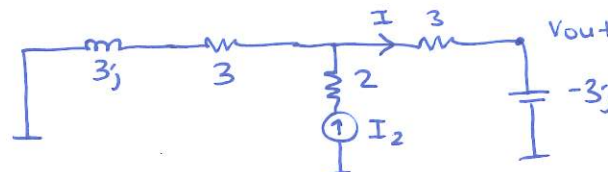


$V_1 = 6\sqrt{2} \angle \pi/4$

$V_1 = 6 + 6j$

$V_{out,1} = V_1 \cdot \frac{-3j}{3j + 3 + 3 - 3j} = (6 + 6j) \cdot \frac{-3j}{6} = \boxed{3 - 3j}$

* Due to I_2 :



$I_2 = 2 \angle -\pi/2$

$I_2 = -2j$

(from current divider)

$I = I_2 \cdot \frac{3 + 3j}{3 + 3j + 3 - 3j} = (-2j) \cdot \frac{3 + 3j}{6} = 1 - j$

$V_{out,2} = I \cdot (-3j) = (1 - j)(-3j) = \boxed{-3 - 3j}$

* So,

$V_{out} = V_{out,1} + V_{out,2} = 3 - 3j - 3 - 3j = -6j = \boxed{6 \angle -\pi/2}$

$= \boxed{6.0 \angle -1.6}$

in two significant digits

(b) [5 points] What is $v_{out}(t)$ in time domain?

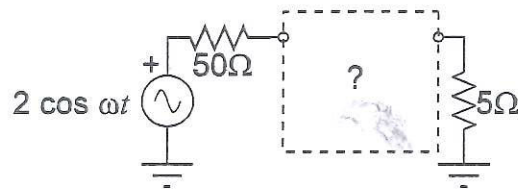
$V_{out} = 6 \angle -\pi/2 \Rightarrow v_{out}(t) = 6 \cos(\omega t - \pi/2) \text{ Volts}$
 $= 6 \sin(\omega t) \text{ Volts}$

(a) $V_{out} = 6 \angle -\pi/2 = 6.0 \angle -1.6$

(b) $v_{out}(t) = 6 \cdot \sin(\omega t) \text{ Volts}$

↑
in two significant digits

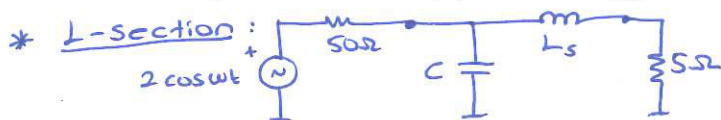
Question 4 [25 Points]:



We have a source with a source impedance of 50Ω , and the load is 5Ω . Design two different circuits between the source and load (in place of the question mark) in such a way that the power delivered to load is maximized at 2 MHz.

Important: If you prefer to use inductors/transformers give the number of turns, using a core with $A_L = 6 \text{ nH/turns}^2$. Use the closest integer value for the number of turns.

(a) [10 points] Your first circuit design. Draw the entire circuit. Give values of all components. If you used inductor(s)/transformer, give the number of turns.

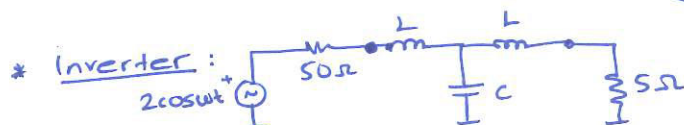


$$50 = (1 + Q^2) 5 \Rightarrow Q = 3$$

$$Q = \frac{\omega_0 L_s}{R_s} \Rightarrow L_s = \frac{5 \cdot 3}{2\pi \cdot 2 \cdot 10^6} = \boxed{1.19 \mu\text{H}}$$

$$L_s = n^2 A_L \Rightarrow n = 14.1 \Rightarrow \boxed{n=14}$$

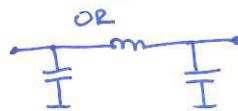
$$C = \frac{1}{\omega_0^2 L_s (1 + \frac{1}{Q^2})} = \boxed{4.8 \text{ nF}}$$



$$x^2 = 5 \cdot 50 \Rightarrow x = \sqrt{250} = 15.8$$

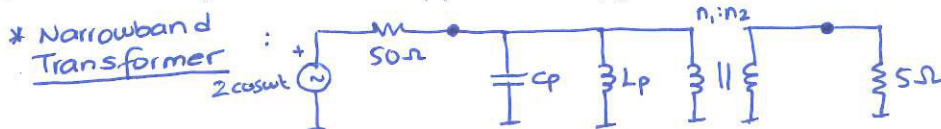
$$C = \frac{1}{\omega_0 \cdot x} = \boxed{5 \text{ nF}}$$

(T-section
or
Π-section
give the same values)



$$L = \frac{x}{\omega_0} = \boxed{1.26 \mu\text{H}} \Rightarrow L = n^2 A_L \Rightarrow n = 14.47 \Rightarrow \boxed{n=14}$$

(b) [10 points] Your second circuit design. Draw the entire circuit. Give values of all components. If you used inductor(s)/transformer, give the number of turns.

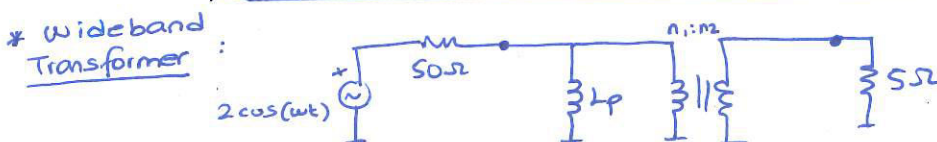


$$\frac{n_1}{n_2} = \sqrt{\frac{50}{5}} \approx 3.16$$

$$\boxed{n_1=19, n_2=6} \text{ gives } \frac{n_1}{n_2} = 3.17$$

$$L_p = n_1^2 A_L = \boxed{2.17 \mu\text{H}}, \quad C_p = \frac{1}{\omega_0^2 L_p} = \boxed{2.9 \text{ nF}}$$

$$L_s = n_2^2 A_L = \boxed{0.21 \mu\text{H}}$$



$$L_p > 4 \left(\frac{n_1}{n_2} \right)^2 \cdot \frac{R_s}{\omega_0}$$

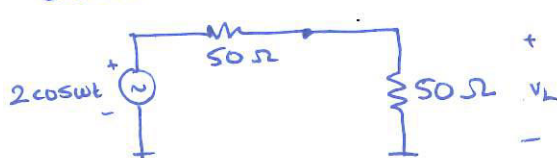
$$L_p > 4 \cdot \frac{50}{2\pi \cdot 2 \cdot 10^6} = 15.9 \mu\text{H}$$

$$n_1 = \sqrt{\frac{L_p}{A_L}} > 51.5 \Rightarrow \boxed{n_1=54, n_2=17}$$

$$\text{gives } \frac{n_1}{n_2} = 3.17$$

(c) [5 points] What is the power delivered to the load (5Ω) for the maximum power transfer case?

Overall circuit is:



load is transformed
to 50Ω

$$P = \frac{1}{2} \frac{V_L^2}{R} = \frac{1}{2} \frac{\left(\frac{2 \cdot 50}{50+50} \right)^2}{50}$$

$$P = \frac{1}{100} \text{ Watt} = \boxed{10 \text{ mW}}$$

$$L_p = n_1^2 A_L = \boxed{17.5 \mu\text{H}}$$

$$L_s = n_2^2 A_L = \boxed{1.7 \mu\text{H}}$$